Area 2: Mechanics of Materials

PE Mechanical: Machine Design & Materials (2025 specification)

Learn each subsection, then take its quiz in the simulator.

Original study material. Not NCEES, Testmasters or PPI content. ★ = focus area.

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2.A Axial Loading

1. Key concepts

- Normal stress σ = P/A acts on a section perpendicular to the load. Elongation δ = PL/(AE) for each uniform segment; add the segments of a stepped bar.
- Statically indeterminate bars: write equilibrium, then compatibility (equal deformations, or deformations that sum to a gap), then the force–deformation relations.
- Parallel members sharing a rigid plate split the load in proportion to their axial stiffness k = AE/L.
- Poisson effect: lateral strain = −ν × axial strain. Diameter change Δd = −ν·ε·d.

2. Essential formulas

Name	Formula	Variables / units
Axial stress	$\sigma = P/A$	psi, MPa
Elongation	$\delta = PL/(AE)$; $\delta_{total} = \Sigma\, P_i L_i/(A_i E_i)$	in, mm
Axial stiffness	$k = AE/L$	
Load sharing (parallel)	$P_1 = P{\cdot}k_1/(k_1{+}k_2)$	equal strain
Poisson	$\Delta d = -\nu(\sigma/E)d$	

3. Handbook search keywords

axial deformation PL/AE · statically indeterminate · compatibility · Poisson's ratio

4. Worked examples

Example 1: Stepped bar
A steel bar (E = 30×10⁶ psi) has 12 in of Ø1.5 in and 8 in of Ø1.0 in and carries 20,000 lb tension. Find the total elongation.

1. A_1 = 1.7671 in², A_2 = 0.7854 in².
2. δ = (20,000/30×10⁶)(12/1.7671 + 8/0.7854) = 0.01132 in.

Answer: 0.0113 in

Example 2: Steel and aluminum posts
A rigid cap sits on a steel post (A = 1.0 in², E = 30×10⁶) and an aluminum post (A = 1.5 in², E = 10×10⁶) of equal length and carries 30,000 lb. Find the steel load.

1. Compatibility: equal shortening, so the load splits in proportion to AE.
2. P_s = 30,000·(30)/(30 + 15) = 20,000 lb; P_a = 10,000 lb.

Answer: 20,000 lb

5. Common traps

- Using one area for a stepped bar.
- Forgetting compatibility in indeterminate problems.
- Unit slips: kN with mm and GPa (1 GPa = 1,000 N/mm²).

2.A Axial Loading: one-page cheat summary

► $\sigma = P/A$
► $\delta = \Sigma PL/AE$
► $k = AE/L$
► Parallel: $P \propto AE/L$
► $\Delta d = -\nu(\sigma/E)d$

Formulas

Axial stress	$\sigma = P/A$
Elongation	$\delta = PL/(AE)$; $\delta_total = \Sigma\ P_iL_i/(A_iE_i)$
Axial stiffness	$k = AE/L$
Load sharing (parallel)	$P_1 = P{\cdot}k_1/(k_1+k_2)$
Poisson	$\Delta d = -\nu(\sigma/E)d$

Search keywords

axial deformation PL/AE, statically indeterminate, compatibility, Poisson's ratio

Top traps

- Using one area for a stepped bar.
- Forgetting compatibility in indeterminate problems.
- Unit slips: kN with mm and GPa (1 GPa = 1,000 N/mm²).

2.B Shear and Transverse Loading

1. Key concepts

- Direct shear (pins, bolts, keys): τ = V/A_shear. Single shear uses 1 area per pin; double shear uses 2.
- Beam transverse shear: τ = VQ/(Ib). It peaks at the neutral axis and is zero at the top and bottom fibers.
- Rectangle: τ_max = 1.5V/A. Solid circle: τ_max = (4/3)V/A. Thin-walled tube: τ_max ≈ 2V/A.
- Shear flow q = VQ/I (force per length) sizes the nails, bolts or welds that join built-up beam parts.

2. Essential formulas

Name	Formula	Variables / units
Direct shear	$\tau = V/(nA)$	n = number of shear planes
Transverse shear	$\tau = VQ/(Ib)$	Q = first moment of the area beyond the cut
Rectangle max	$\tau_{max} = 3V/(2A)$	
Circle max	$\tau_{max} = 4V/(3A)$	
Shear flow	$q = VQ/I$; fastener spacing $s = F_{allow}/q$	lb/in

3. Handbook search keywords

transverse shear stress VQ/Ib · shear flow · double shear · first moment of area

4. Worked examples

Example 1: Rectangular beam
A 3 in × 8 in rectangular beam carries V = 9,600 lb. Find τ_max.

1. A = 24 in².
2. τ_max = 1.5(9,600)/24 = 600 psi at the neutral axis.

Answer: 600 psi

Example 2: Clevis pin in double shear
A Ø0.625 in pin carries 8,000 lb in double shear. Find τ.

1. A = π(0.625)²/4 = 0.3068 in².
2. τ = 8,000/(2 × 0.3068) = 13,038 psi.

Answer: 13,038 psi

5. Common traps

- Using V/A (the average) when the maximum is asked.
- Forgetting double shear (2 planes).
- Using 4/3 for a rectangle or 3/2 for a circle.

2.B Shear and Transverse Loading: one-page cheat summary

- ► $\tau = V/(nA)$
- ► $\tau = VQ/Ib$
- ► Rect $1.5V/A$; circle $4V/3A$; tube $2V/A$
- ► $q = VQ/I$

Formulas

Direct shear	$\tau = V/(nA)$
Transverse shear	$\tau = VQ/(Ib)$
Rectangle max	$\tau_{max} = 3V/(2A)$
Circle max	$\tau_{max} = 4V/(3A)$
Shear flow	$q = VQ/I$; fastener spacing $s = F_{allow}/q$

Search keywords

transverse shear stress VQ/Ib, shear flow, double shear, first moment of area

Top traps

- Using V/A (the average) when the maximum is asked.
- Forgetting double shear (2 planes).
- Using 4/3 for a rectangle or 3/2 for a circle.

2.C Bending

1. Key concepts

- Draw shear and moment diagrams. Shear is the slope of the moment curve, so moment peaks where V = 0. Jumps happen at point loads (V) and applied couples (M).
- Flexure formula: σ = Mc/I = M/S, with tension and compression on opposite sides of the neutral axis.
- Parallel axis: I = I_c + Ad². For a composite section, find the centroid first.
- Deflection: use the handbook beam tables and superpose load cases. Simply supported with a center load: PL³/48EI. Cantilever with an end load: PL³/3EI. UDL on a simple span: 5wL⁴/384EI.

2. Essential formulas

Name	Formula	Variables / units
Flexure	$\sigma = Mc/I = M/S$	S = I/c
Rectangle	$I = bh^3/12$; $S = bh^2/6$	
Circle	$I = \pi d^4/64$; $S = \pi d^3/32$	
Parallel axis	$I = I_c + Ad^2$	
SS center load	$M_{max} = PL/4$; $\delta = PL^3/48EI$	
SS UDL	$M_{max} = wL^2/8$; $\delta = 5wL^4/384EI$	
Cantilever end load	$M_{max} = PL$; $\delta = PL^3/3EI$	
Cantilever UDL	$M_{max} = wL^2/2$; $\delta = wL^4/8EI$	

3. Handbook search keywords

flexure formula Mc/I | shear and moment diagrams | beam deflection tables | section modulus | parallel axis theorem

4. Worked examples

Example 1: Wood beam with UDL
A 12 ft simple span carries 400 lb/ft. The section is 3.5 in × 9.25 in. Find σ_max.

1. M = wL²/8 = 400(12)²/8 = 7,200 ft-lb = 86,400 in-lb.
2. S = bh²/6 = 49.9 in³.
3. σ = M/S = 1,731 psi.

Answer: 1,731 psi

Example 2: Midspan deflection
A simply supported steel beam (E = 29×10⁶ psi, I = 100 in⁴) spans 120 in with 10,000 lb at midspan. Find δ.

1. δ = PL³/48EI = 10,000(120)³/(48·29×10⁶·100) = 0.1241 in.

Answer: 0.1241 in

5. Common traps

- Leaving the moment in ft-lb while S is in in³ (multiply by 12).
- Using c = h instead of h/2.
- Taking the moment at the wrong location (max M is where V = 0).
- Mixing the cantilever and simply supported formulas.

2.C Bending: one-page cheat summary

- ▶ $\sigma = Mc/I = M/S$
- ▶ Rect $S = bh^2/6$
- ▶ Max M where $V = 0$
- ▶ SS: $PL/4, wL^2/8$
- ▶ δ: $PL^3/48EI, 5wL^4/384EI, PL^3/3EI, wL^4/8EI$
- ▶ $I = I_c + Ad^2$

Formulas

Flexure	$\sigma = Mc/I = M/S$
Rectangle	$I = bh^3/12$; $S = bh^2/6$
Circle	$I = \pi d^4/64$; $S = \pi d^3/32$
Parallel axis	$I = I_c + Ad^2$
SS center load	$M_{max} = PL/4$; $\delta = PL^3/48EI$
SS UDL	$M_{max} = wL^2/8$; $\delta = 5wL^4/384EI$
Cantilever end load	$M_{max} = PL$; $\delta = PL^3/3EI$
Cantilever UDL	$M_{max} = wL^2/2$; $\delta = wL^4/8EI$

Search keywords

flexure formula Mc/I, shear and moment diagrams, beam deflection tables, section modulus, parallel axis theorem

Top traps

- Leaving the moment in ft-lb while S is in in^3 (multiply by 12).
- Using c = h instead of h/2.
- Taking the moment at the wrong location (max M is where V = 0).

2.D Buckling

1. Key concepts

- Long slender columns fail by elastic buckling (Euler). Short columns fail by yielding. Intermediate columns follow the Johnson parabola.
- Effective length L_e = KL. Theoretical K: fixed–fixed 0.5, fixed–pinned 0.7, pinned–pinned 1.0, fixed–free 2.0. Design (recommended) K values are a little higher.
- Slenderness ratio = KL/r, with r = √(I/A). Compare it with the transition value C_c = √(2π²E/Sy). Above C_c use Euler; below it use Johnson.
- Always buckle about the weak axis (smallest I).

2. Essential formulas

Name	Formula	Variables / units
Euler load	$P_{cr} = \pi^2 EI/(KL)^2$	
Euler stress	$\sigma_{cr} = \pi^2 E/(KL/r)^2$	
Radius of gyration	$r = \sqrt{(I/A)}$; solid round $r = d/4$	
Transition	$C_c = \sqrt{(2\pi^2 E/Sy)}$	
Johnson	$P_{cr} = A[Sy - (Sy \cdot KL/r)^2/(4\pi^2 E)]$	intermediate

3. Handbook search keywords

Euler buckling · effective length factor · slenderness ratio · Johnson formula · radius of gyration

4. Worked examples

Example 1: Pinned steel column
A Ø1.25 in solid steel rod (E = 29×10⁶ psi, Sy = 50 ksi), 60 in long and pinned at both ends. Find P_cr.

1. r = d/4 = 0.3125 in; KL/r = 60/0.3125 = 192.
2. C_c = √(2π²·29×10⁶/50,000) = 107. Since 192 > 107, use Euler.
3. I = πd⁴/64 = 0.1198 in⁴; P_cr = π²EI/L² = 9,528 lb.

Answer: 9,528 lb

5. Common traps

- Using Euler for a short column (check KL/r against C_c).
- Using the strong-axis I.
- Swapping K for fixed–free (2.0) and fixed–fixed (0.5).

2.D Buckling: one-page cheat summary

- ► $P_{cr} = \pi^2 EI/(KL)^2$
- ► K: 0.5 / 0.7 / 1 / 2
- ► $r = \sqrt{(I/A)}$, round $d/4$
- ► $C_c = \sqrt{(2\pi^2 E/Sy)}$
- ► Johnson below C_c

Formulas

Euler load	$P_{cr} = \pi^2 EI/(KL)^2$
Euler stress	$\sigma_{cr} = \pi^2 E/(KL/r)^2$
Radius of gyration	$r = \sqrt{(I/A)}$; solid round $r = d/4$
Transition	$C_c = \sqrt{(2\pi^2 E/Sy)}$
Johnson	$P_{cr} = A[Sy - (Sy \cdot KL/r)^2/(4\pi^2 E)]$

Search keywords

Euler buckling, effective length factor, slenderness ratio, Johnson formula, radius of gyration

Top traps

- Using Euler for a short column (check KL/r against C_c).
- Using the strong-axis I.
- Swapping K for fixed–free (2.0) and fixed–fixed (0.5).

2.E Torsion

1. Key concepts

- Circular shafts: τ = Tc/J peaks at the outer surface. Angle of twist θ = TL/(JG) in radians.
- Power and torque: T(in-lb) = 63,025·hp/rpm; T(ft-lb) = 5,252·hp/rpm; T(N·m) = 9,549·kW/rpm.
- Hollow shafts: J = π(D⁴ − d⁴)/32. For the same weight they carry more torque.
- Thin-walled closed tube: τ = T/(2A_m t), where A_m is the area enclosed by the mid-wall line.

2. Essential formulas

Name	Formula	Variables / units
Torsion stress	$\tau = Tc/J = 16T/(\pi d^3)$ (solid)	
Polar moment	$J = \pi d^4/32$; hollow $\pi(D^4-d^4)/32$	
Twist	$\theta = TL/(JG)$ [rad]	G steel ≈ 11.5×10⁶ psi = 80 GPa
Power	$T = 63{,}025\ hp/n$ (in-lb)	
Thin tube	$\tau = T/(2A_m\ t)$	

3. Handbook search keywords

torsion shear stress Tc/J · angle of twist · polar moment of inertia · power torque speed

4. Worked examples

Example 1: Size a shaft for torque
A shaft transmits 20 hp at 600 rpm with an allowable τ of 12,000 psi. Find the minimum solid diameter.

1. T = 63,025(20)/600 = 2,101 in-lb.
2. d = (16T/πτ)^(1/3) = 0.962 in.

Answer: 0.962 in (use the next standard size)

Example 2: Twist
Same torque on a Ø1.25 in shaft, 48 in long, with G = 11.5×10⁶ psi. Find θ.

1. J = π(1.25)⁴/32 = 0.2397 in⁴.
2. θ = TL/JG = 0.0366 rad = 2.10°.

Answer: 2.10°

5. Common traps

- Leaving θ in radians when degrees are asked.
- Using I (πd⁴/64) instead of J.
- Using 5,252 (gives ft-lb) and then using in-lb.

2.E Torsion: one-page cheat summary

- ► $\tau = 16T/\pi d^3$
- ► $J = \pi d^4/32$
- ► $\theta = TL/JG$ (rad)
- ► $T = 63{,}025\ hp/n$ in-lb
- ► Thin tube $\tau = T/2A_m t$

Formulas

Torsion stress	$\tau = Tc/J = 16T/(\pi d^3)$ (solid)
Polar moment	$J = \pi d^4/32$; hollow $\pi(D^4 - d^4)/32$
Twist	$\theta = TL/(JG)$ [rad]
Power	$T = 63{,}025\ hp/n$ (in-lb)
Thin tube	$\tau = T/(2A_m t)$

Search keywords

torsion shear stress Tc/J, angle of twist, polar moment of inertia, power torque speed

Top traps

- Leaving θ in radians when degrees are asked.
- Using I ($\pi d^4/64$) instead of J.
- Using 5,252 (gives ft-lb) and then using in-lb.

2.F Combined Loading

1. Key concepts

- Superpose stresses at one point: axial P/A, bending ±Mc/I, torsion Tc/J, transverse VQ/Ib. Find the critical point (usually the outer fiber).
- Mohr's circle: center C = (σx+σy)/2, radius R = √[((σx−σy)/2)² + τ²]. $\sigma_{1,2}$ = C ± R and τ_max,in-plane = R.
- In 3-D, the absolute maximum shear = (σ_max − σ_min)/2, counting σ_3 = 0 in plane stress.
- Shaft with bending plus torsion: σ = 32M/πd³, τ = 16T/πd³, von Mises σ′ = √(σ² + 3τ²).

2. Essential formulas

Name	Formula	Variables / units
Principal stresses	$\sigma_{1,2} = (\sigma x+\sigma y)/2 \pm \sqrt{[((\sigma x-\sigma y)/2)^2 + \tau xy^2]}$	
Max in-plane shear	$\tau_max = \sqrt{[((\sigma x-\sigma y)/2)^2 + \tau xy^2]}$	
Principal angle	$\tan 2\theta p = 2\tau xy/(\sigma x-\sigma y)$	
Combined axial + bending	$\sigma = P/A \pm Mc/I$	
Shaft	$\sigma' = \sqrt{(\sigma^2 + 3\tau^2)}$	

3. Handbook search keywords

Mohr's circle · principal stresses · combined stresses · von Mises

4. Worked examples

Example 1: Principal stresses
σx = 12 ksi, σy = −4 ksi, τxy = 6 ksi. Find σ_1, σ_2 and τ_max.

1. C = 4 ksi; R = √(8² + 6²) = 10 ksi.
2. σ_1 = 14 ksi; σ_2 = -6 ksi; τ_max = 10 ksi.

Answer: σ_1 = 14, σ_2 = −6, τ_max = 10 ksi

5. Common traps

- Forgetting σ_3 = 0 when σ_1 and σ_2 have the same sign (then τ_abs = σ_1/2).
- Sign errors for compressive σy.
- Using √(σ² + 4τ²) (Tresca) when von Mises is asked.

2.F Combined Loading: one-page cheat summary

- ► $C = (\sigma x + \sigma y)/2$
- ► $R = \sqrt{[((\sigma x - \sigma y)/2)^2 + \tau^2]}$
- ► $\sigma_{1,2} = C \pm R$
- ► $\tau_{max} = R$ (in-plane)
- ► $\sigma' = \sqrt{(\sigma^2 + 3\tau^2)}$

Formulas

Principal stresses	$\sigma_{1,2} = (\sigma x + \sigma y)/2 \pm \sqrt{[((\sigma x - \sigma y)/2)^2 + \tau xy^2]}$
Max in-plane shear	$\tau_{max} = \sqrt{[((\sigma x - \sigma y)/2)^2 + \tau xy^2]}$
Principal angle	$\tan 2\theta p = 2\tau xy/(\sigma x - \sigma y)$
Combined axial + bending	$\sigma = P/A \pm Mc/I$
Shaft	$\sigma' = \sqrt{(\sigma^2 + 3\tau^2)}$

Search keywords

Mohr's circle, principal stresses, combined stresses, von Mises

Top traps

- Forgetting $\sigma_3 = 0$ when σ_1 and σ_2 have the same sign (then $\tau_{abs} = \sigma_1/2$).
- Sign errors for compressive σy.
- Using $\sqrt{(\sigma^2 + 4\tau^2)}$ (Tresca) when von Mises is asked.

2.G Static Failure

1. Key concepts

- Ductile materials: use distortion energy (von Mises, the best fit) or maximum shear stress (Tresca, conservative). Brittle materials: use maximum normal stress, Coulomb–Mohr or modified Mohr.
- von Mises plane stress: $\sigma' = \sqrt{(\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2)} = \sqrt{(\sigma x^2 - \sigma x \sigma y + \sigma y^2 + 3\tau^2)}$. n = Sy/σ′.
- Tresca: n = Sy/(σ_max − σ_min), including $\sigma_3 = 0$. Shear yield: von Mises Ssy = 0.577Sy; Tresca 0.5Sy.
- Static stress concentration: ductile materials can usually ignore Kt (local yielding relieves it). Brittle materials must apply Kt.

2. Essential formulas

Name	Formula	Variables / units
von Mises	$\sigma' = \sqrt{(\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2)}$; $n = Sy/\sigma'$	
Tresca	$n = Sy/(\sigma_1 - \sigma_3)$	
Shear yield	$Ssy = 0.577Sy\ (DE), 0.5Sy\ (MSS)$	
Max normal (brittle)	$n = Sut/\sigma_1$	

3. Handbook search keywords

distortion energy theory · maximum shear stress theory · factor of safety · Mohr theory brittle

4. Worked examples

Example 1: von Mises vs Tresca
Use $\sigma_1 = 14$ ksi and $\sigma_2 = -6$ ksi from the 2.F example, with Sy = 50 ksi. Find n by both theories.

1. $\sigma' = \sqrt{(14^2 - 14(-6) + 6^2)} = 17.78$ ksi → n = 2.81.
2. Tresca: $\sigma_1 - \sigma_3 = 14 - (-6) = 20$ ksi → n = 2.50.

Answer: n_DE = 2.81; n_Tresca = 2.50

5. Common traps

- Using Tresca with $\sigma_1 - \sigma_2$ when both are positive ($\sigma_3 = 0$ governs).
- Applying Kt to a ductile part under static load.
- Applying the brittle theories to steel.

2.G Static Failure: one-page cheat summary

- ► $\sigma' = \sqrt{(\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2)}$
- ► Tresca $n = Sy/(\sigma max - \sigma min)$
- ► Ssy: $0.577Sy$ / $0.5Sy$
- ► Brittle: Coulomb–Mohr
- ► Ductile static: ignore Kt

Formulas

von Mises	$\sigma' = \sqrt{(\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2)}$; $n = Sy/\sigma'$
Tresca	$n = Sy/(\sigma_1 - \sigma_3)$
Shear yield	$Ssy = 0.577Sy$ (DE), $0.5Sy$ (MSS)
Max normal (brittle)	$n = Sut/\sigma_1$

Search keywords

distortion energy theory, maximum shear stress theory, factor of safety, Mohr theory brittle

Top traps

- Using Tresca with $\sigma_1 - \sigma_2$ when both are positive ($\sigma_3 = 0$ governs).
- Applying Kt to a ductile part under static load.
- Applying the brittle theories to steel.

2.H Fatigue Failure

1. Key concepts

- The endurance limit of steel: Se′ ≈ 0.5Sut (Sut ≤ 200 ksi), or 100 ksi above that. Correct it with the Marin factors Se = ka·kb·kc·kd·ke·Se′ (surface, size, load, temperature, reliability).
- Fluctuating stress: σa = (σmax − σmin)/2, σm = (σmax + σmin)/2. Apply K_f = 1 + q(Kt − 1) to the stresses.
- Mean-stress criteria: modified Goodman 1/n = σa/Se + σm/Sut (conservative, the usual exam choice). Gerber is a parabola (better fit). ASME-elliptic uses Sy. Also check first-cycle yield: σa + σm ≤ Sy/n.
- Finite life: S-N line through (10^3, f·Sut) and (10^6, Se), with S = aN^b. Miner's rule: $\Sigma n_i/N_i = 1$.

2. Essential formulas

Name	Formula	Variables / units
Endurance limit	$Se' = 0.5Sut$ (steel)	
Surface factor	$ka = a \cdot Sut^b$ (machined: a = 2.70, b = −0.265 ksi)	
Size factor	$kb = 0.879d^{-0.107}$ (0.11–2 in)	
Fatigue Kt	$K_f = 1 + q(Kt - 1)$	
Goodman	$1/n = \sigma a/Se + \sigma m/Sut$	
Basquin	$S_f = aN^b$; $b = -(1/3)\log(f Sut/Se)$; $a = (f Sut)^2/Se$	
Miner	$\Sigma\, n_i/N_i = 1$	

3. Handbook search keywords

endurance limit · Marin factors · modified Goodman · notch sensitivity · S-N diagram · Miner's rule

4. Worked examples

Example 1: Goodman factor of safety
A machined Ø1.0 in steel shaft (Sut = 90 ksi) sees σa = 18 ksi and σm = 20 ksi (K_f already applied), in rotating bending. Find n by Goodman.

1. ka = 2.70(90)^−0.265 = 0.819; kb = 0.879(1.0)^−0.107 = 0.879.
2. Se = 0.5(90)(0.819)(0.879) = 32.4 ksi.
3. 1/n = 18/32.4 + 20/90 → n = 1.29.

Answer: n ≈ 1.29

Example 2: Fatigue stress concentration
Kt = 2.4 and q = 0.85. Find K_f.

1. K_f = 1 + 0.85(1.4) = 2.19.

Answer: 2.19

5. Common traps

- Forgetting the Marin factors (using the raw 0.5Sut).
- Applying K_f to the strength instead of the stress (or applying it twice).
- Using σmax instead of σa and σm.
- Skipping the first-cycle yield check.

2.H Fatigue Failure: one-page cheat summary

- ▶ $S_e' = 0.5S_{ut}$
- ▶ $S_e = k_a k_b k_c k_d k_e S_e'$
- ▶ $K_f = 1 + q(K_t - 1)$
- ▶ $\sigma_a = (\max - \min)/2$; $\sigma_m = (\max + \min)/2$
- ▶ Goodman: $1/n = \sigma_a/S_e + \sigma_m/S_{ut}$
- ▶ Miner: $\Sigma n/N = 1$

Formulas

Endurance limit	$S_e' = 0.5S_{ut}$ (steel)
Surface factor	$k_a = aS_{ut}^b$ (machined: a = 2.70, b = −0.265 ksi)
Size factor	$k_b = 0.879d^{-0.107}$ (0.11–2 in)
Fatigue Kt	$K_f = 1 + q(K_t - 1)$
Goodman	$1/n = \sigma_a/S_e + \sigma_m/S_{ut}$
Basquin	$S_f = aN^b$; $b = -(1/3)\log(fS_{ut}/S_e)$; $a = (fS_{ut})^2/S_e$
Miner	$\Sigma\, n_i/N_i = 1$

Search keywords

endurance limit, Marin factors, modified Goodman, notch sensitivity, S-N diagram, Miner's rule

Top traps

- Forgetting the Marin factors (using the raw 0.5Sut).
- Applying K_f to the strength instead of the stress (or applying it twice).
- Using σmax instead of σa and σm.

2.I Thermal and Interference Stresses

1. Key concepts

- A fully restrained bar heated by ΔT: σ = EαΔT (compressive). If there is a gap, only the expansion beyond the gap is restrained.
- Press and shrink fits: the interface pressure p comes from the interference δ (diametral). Same material, solid shaft: p = Eδ(D_o² − d²)/(2dD_o²).
- Hub hoop stress at the bore is σ_t = p(D_o² + d²)/(D_o² − d²). Torque capacity T = πμpd²L/2. Axial force capacity F = πμpdL.
- Shrink-fit assembly: the hub temperature rise is ΔT = (δ + clearance)/(αd).

2. Essential formulas

Name	Formula	Variables / units
Restrained bar	$\sigma = E\alpha\Delta T$	with gap: σ = E(αΔTL − gap)/L
Interference pressure	$p = E\delta(D_o^2 - d^2)/(2dD_o^2)$	same E, solid shaft
Hub bore hoop	$\sigma_t = p(D_o^2 + d^2)/(D_o^2 - d^2)$	
Torque capacity	$T = \pi\mu p d^2 L/2$	
Shrink temp	$\Delta T = (\delta + c)/(\alpha d)$	

3. Handbook search keywords

thermal stress · interference fit pressure · shrink fit · press fit torque

4. Worked examples

Example 1: Restrained rail
A steel bar (E = 29×10⁶ psi, α = 6.5×10⁻⁶/°F) is fixed between walls and heated 100 °F. Find σ.

1. σ = EαΔT = 29×10⁶(6.5×10⁻⁶)(100) = 18,850 psi (compression).

Answer: 18,850 psi

Example 2: Press-fit torque
A Ø2.0 in steel shaft is pressed into a steel hub (OD 4.0 in, L = 2.5 in) with δ = 0.002 in, E = 30×10⁶ psi, μ = 0.15. Find p and T.

1. p = 30×10⁶(0.002)(16 − 4)/(2·2·16) = 11,250 psi.
2. T = π(0.15)p(2²)(2.5)/2 = 26,507 in-lb.

Answer: T ≈ 26,507 in-lb

5. Common traps

- Using a radial interference where the formula expects diametral (or vice versa).
- Ignoring the gap.
- Forgetting that thermal stress does not depend on length when the bar is fully restrained.

2.I Thermal and Interference Stresses: one-page cheat summary

- ► $\sigma = E\alpha\Delta T$
- ► $p = E\delta(D_o^2 - d^2)/(2dD_o^2)$
- ► $T = \pi\mu p d^2 L/2$
- ► $\Delta T_{shrink} = (\delta + c)/(\alpha d)$

Formulas

Restrained bar	$\sigma = E\alpha\Delta T$
Interference pressure	$p = E\delta(D_o^2 - d^2)/(2dD_o^2)$
Hub bore hoop	$\sigma_t = p(D_o^2 + d^2)/(D_o^2 - d^2)$
Torque capacity	$T = \pi\mu p d^2 L/2$
Shrink temp	$\Delta T = (\delta + c)/(\alpha d)$

Search keywords

thermal stress, interference fit pressure, shrink fit, press fit torque

Top traps

- Using a radial interference where the formula expects diametral (or vice versa).
- Ignoring the gap.
- Forgetting that thermal stress does not depend on length when the bar is fully restrained.